

NIKITA BHANSALI

☎ +91-8585804977 ✉ nbhansali2006@gmail.com [in linkedin.com/in/nikitabhansali](https://www.linkedin.com/in/nikitabhansali) github.com/nikita123

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering; **CGPA:8.48**

2023 – 2027

Bhopal, India

SKILL SET

Embedded Systems & Hardware: ESP32/ESP32-CAM, Arduino IDE, Microcontroller Programming, Sensor Interfacing (DHT11, PIR), Relay & Actuator Control, Circuit Design

RF & Signal Processing: Software Defined Radio (RTL-SDR, HackRF), GNU Radio, FFT/Spectrum Analysis, Antenna Design (Yagi-Uda, Helical), RF Synthesizers (ADF4351), CST Studio/HFSS

Languages: C, C++, Python, Java, SQL, HTML/CSS, JavaScript

AI/ML & Dev Tools: Machine Learning, Deep Learning, Computer Vision, TensorFlow, Scikit-learn, Git/GitHub, Django, React Native, MERN Stack

Databases: MySQL, MongoDB

INTERNSHIP

MPOnline (Government of MP) | AI/ML & Advanced Software Engineering Internship

May 2026 – Jul 2026

-Built **ML models** (regression, classification, decision trees, random forest, SVM, clustering, PCA) using **Python, NumPy, Pandas**, and **Scikit-learn**, and developed neural networks with **TensorFlow/PyTorch** for computer vision and NLP.

-Practiced full-stack **SDLC: DBMS, SQL, OOP, Git/GitHub**, and **CRUD** app development, followed by requirement analysis, system design, backend development, testing, and cloud deployment for a capstone project.

PROJECTS

Anti-Drone System using SDR | SDR, RF Systems, Embedded Design

-Real-time anti-drone detection system monitoring the **2.4 GHz/5.8 GHz** ISM bands to identify FHSS control signals from commercial drones, for campus perimeter security and anti-UAV surveillance.

-Built using an **RTL-SDR/HackRF** for spectrum sensing, an **ESP32-S3** for target identification, an **ADF4351** RF synthesizer for interference generation, and directional antennas; implemented in **GNU Radio** and **Python (PySDR)**.

-Researched **FFT**-based signal detection and simulated Yagi-Uda/Helical antennas in **CST Studio/HFSS** to focus jamming energy and minimize side-band interference.

Home Automation with LumaSync | IoT, Embedded Systems, ESP32

-IoT-based home automation and security system regulating room temperature, detecting motion, and capturing images of entrants, controllable via a local web dashboard and Google Assistant.

-Built using an **ESP32/ESP32-CAM**, a **DHT11** sensor, a **PIR** sensor, and a **4-channel relay module**; developed a real-time dashboard (**HTML/CSS/JS**) and integrated **Google Assistant** via **IFTTT**.

-Researched predictive, trend-based temperature control logic (dynamic hysteresis) to reduce overshoot and improve system responsiveness.

Smart Surveillance for Public Transport | Computer Vision, Embedded AI

-Real-time monitoring system for public transport to detect anomalies and enhance passenger safety.

-Built using **OpenCV** and **YOLO** for real-time object detection and video analysis on embedded/edge hardware.

-Researched anomaly-detection models and optimized **YOLO** inference for real-time performance on live feeds.

CERTIFICATIONS

Applied Machine Learning in Python – Coursera (University of Michigan)

[\[Link\]](#)

Agentblazer Champion 2026 – ServiceNow

[\[Link\]](#)

Health Hackathon 2026 – Johns Hopkins University Collaboration – Participation & Project Showcase

[\[Link\]](#)

Health Hackathon 2025 – Johns Hopkins University Collaboration – Participation & Project Showcase

[\[Link\]](#)

EXTRACURRICULAR

Competitive Programming: Solved **250+ problems** on LeetCode; maintaining consistent coding streaks.

[\[Link\]](#)

Open Source: Active contributor to GitHub projects focusing on AI/ML tools and web accessibility.

Languages: Proficient in English, Hindi; learning **Japanese** and **German** (250+ day Duolingo streak).

[\[Link\]](#)

Leadership: Student Partner at **IEEE** and **VITronix Club**, organizing technical workshops for 1000+ students.